

HEREDITY IN ORGANISMS

In your previous class you have already learnt about structure of animal and plant cell. You know that cells are the basic functional and structural unit of living things. You have also seen that living things undergo growth and development process. Do you know how you and other living things have developed and grown over the years? If you burn yourself, then how do your wounds heal? How do living things reproduce? Why children resemble their parents? You may even have some resemblance with your maternal or parental uncles and aunts or grandparents. Which structure in cells helps living things to grow in the same manner as their parents grow? How characteristics are transferred from parent to offspring? Let us explore these and other related questions.

In this Chapter you will learn about:

- Cell Division.
- Heredity.
- Basis of Heredity (Chromosome, DNA and Genes in Plant and Animal Cell).

All the students will be able to:

- ✓ Differentiate between Mitosis and Meiosis.
- ✓ Identify DNA and chromosomes in the cell diagram.
- ✓ Define heredity and recognize its importance in transferring of characteristics from parents to offsprings.
- ✓ Identify the characteristics that can be transferred from parent to offsprings.
- ✓ Compare characteristics related to ear and eye colour.

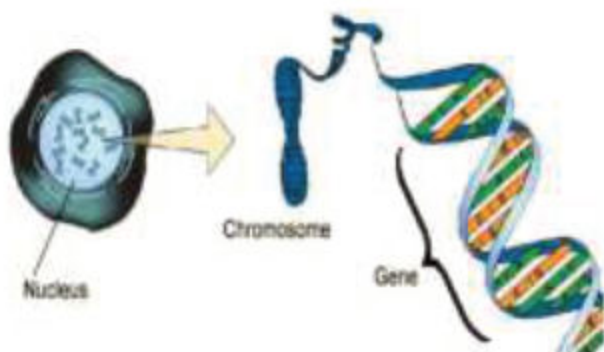


Figure 2.1: Nucleus, Chromosomes and Genes

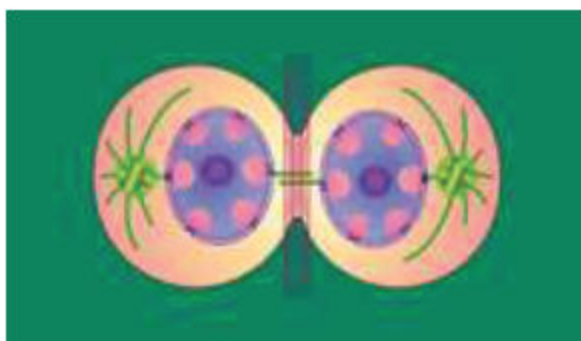


Figure 2.2: Cell Division

CELL DIVISION

➤ Differentiate between Mitosis and Meiosis.

You know that all multicellular organisms (plants, animals and human beings) are composed of more than a billion cells which are the structural and functional unit of life. Cells are continuously dying and are replaced by new cells. The multiplication of new cells takes place by a process known as cell division. The cell division involves nuclear division and cytoplasmic division. The nuclear division is called Karyokinesis, followed by

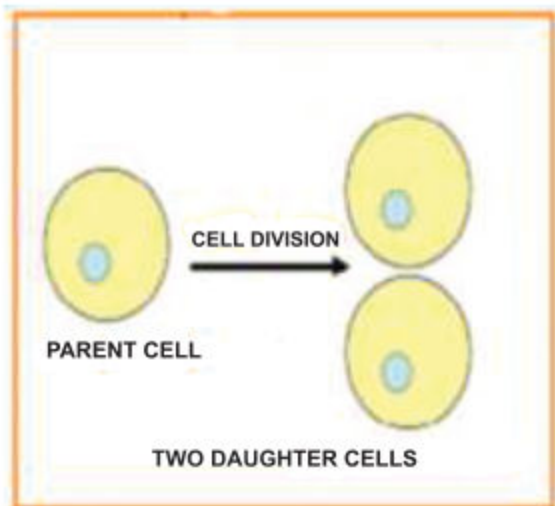


Fig. 2.3: Cell Division

division of cytoplasm called cytokinesis. The nucleus in a cell takes part in cell division. The nucleus contains a hereditary material called chromosomes which divides and form new cells. The cell that divides is called **parent cell** resulting in producing new cells called **daughter cells**. Before the cell division, the dividing cell undergoes a phase called interphase in which the sets of chromosomes are duplicated in the parent cells.

Cell division takes place mainly by two processes:

1. **Mitosis** in which body cells divide for growth, repairing and development.
2. **Meiosis** takes place when sex cells are divided and gametes are formed during the process of reproduction in humans, other

DO YOU KNOW?

- Nucleus is responsible for cell division because it contains nuclear material called chromosomes.
- Chromosomes are made up of proteins and nucleic acid.
- Famous Scientists and Nobel laureate Francis Crick and James Watson presented the first DNA model in 1953.
- For any organisms, the number of chromosomes present in the cell remains constant. The number of chromosomes present in a human cell is 46.

A. MITOSIS

Mitosis takes place in somatic or vegetative cells as a normal process of growth and development. In mitosis, the parent cell divides into two daughter cells with the exact number of chromosomes which are the bearers of the hereditary characteristics. Genes which are responsible for the production of characteristics are found in the chromosomes. Prior to mitosis, interphase occurs as part of cell division. Mitosis consists of four stages **prophase**, **metaphase**, **anaphase** and **telophase**.

- I. **Prophase** is the first stage when chromatin become thickened, shorten and become visible under a microscope called chromosomes.
- ii. During **Metaphase**, chromosomes get attached to spindle fibres formed by the two centrioles.
- iii. In **anaphase**, contraction of spindle fibre takes place resulting the chromosomes to move towards the opposite poles of the cells.
- iv. **Telophase** is the last stage where chromosomes reach towards their respective poles and followed by the process of cytokinesis resulting in forming two daughter cells. The daughter cells are the exact copy of parent cells with same number of chromosomes.

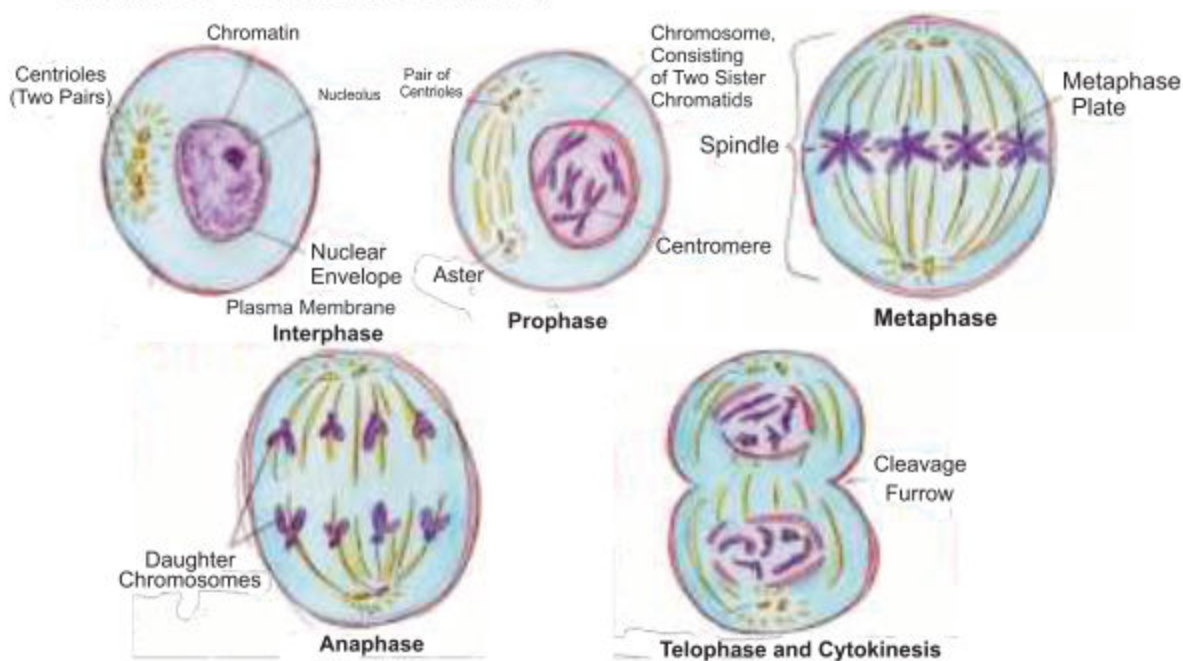


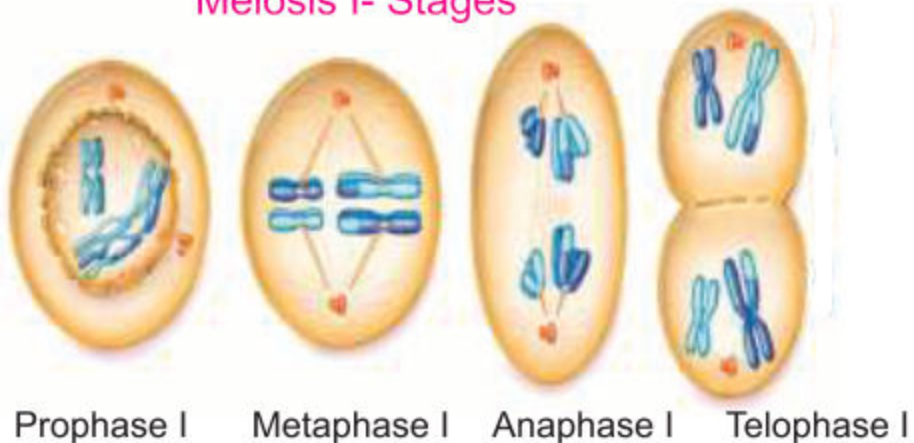
Fig.2.4: Phases of Mitosis

B. MEIOSIS

Meiosis is also known as a reduction cell division. Meiosis consists of two nuclear divisions, Meiosis I and Meiosis II which are needed for sexual reproduction. In meiosis, the daughter cells produced have half the number of chromosomes (haploid) than the parent cells. Meiosis I is a reduction division in which the number of chromosomes is reduced.

However, meiosis II is similar to mitotic division. The number of chromosome remain the same (half) during the formation of four daughter cells. It occurs in sex organs during gamete formation.

Meiosis I- Stages



Meiosis II- Stages

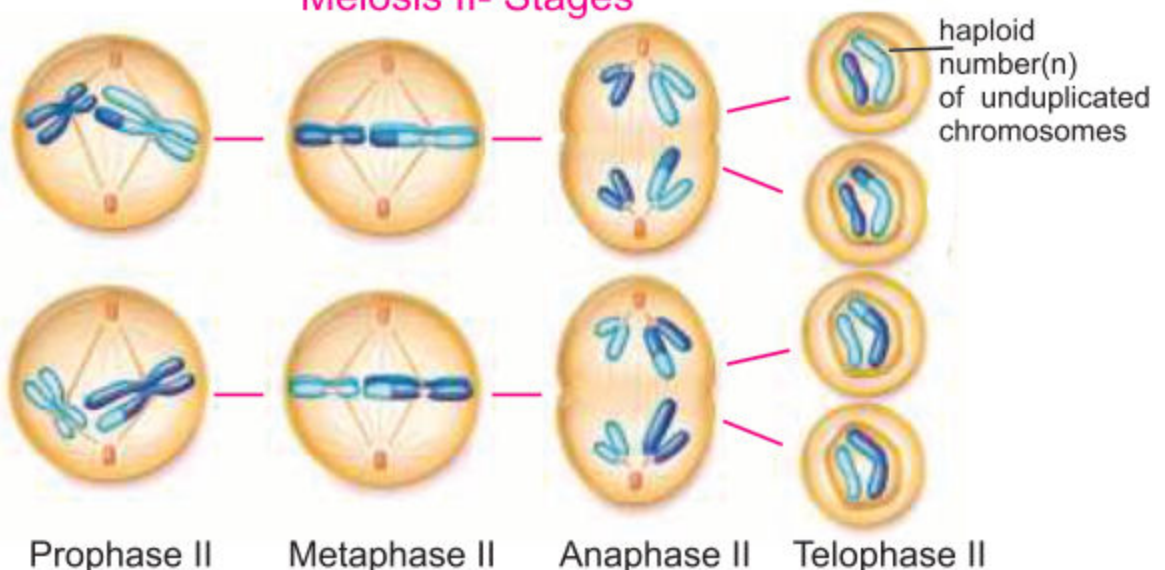


Fig.2.5: Phases of Meiosis I and Meiosis II

Difference between Mitosis and Meiosis.

Mitosis	Meiosis
1. It occurs in somatic (body) cells, when parent cells divide during growth, repair and development.	It is a special type of cell division necessary for sexual reproduction. It occurs in sex organs during formation of gamete (egg/sperm, spores).
2. In mitosis, two identical daughter cells having same number of chromosomes out of one parent cell are produced.	In meiosis, four daughter cells are produced containing haploid number of chromosomes.
3. Mitosis consists of one round.	Meiosis consists of two rounds, Meiosis I and Meiosis II.
4. Cytokinesis occurs just after Telophase which is the last stage of Mitosis.	Cytokinesis or division of cytoplasm takes place in both Meiosis I and Meiosis II.
5. Chromosome number remains same.	Chromosomes number become half (haploid).

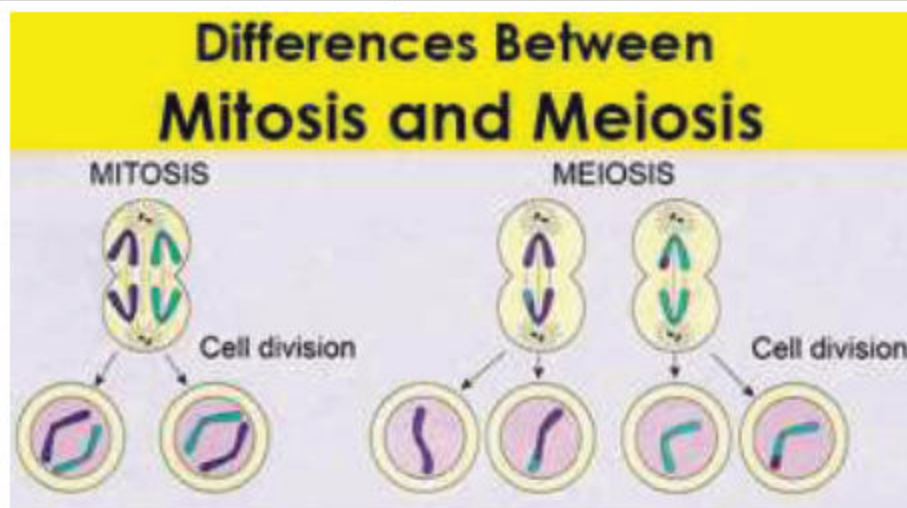


Fig. 2.6: Difference between Mitosis and Meiosis

Teacher Note:

Teacher should show prepared slides of each phase of mitosis and meiosis under a microscope and ask students to observe and draw the diagrams of every stage in their exercise books. Teacher should ensure that each student observe, draws the diagrams and discusses the difference between Mitosis and Meiosis.

Making Model of Mitosis and Meiosis

Activity 2.1:

Making Model of Mitosis and Meiosis

Material Required:

- White disposable plates, 12.
- Threads of two different colours.
- Beads.
- Markers.
- Scissors.



Method:

You may use the white plate to represent the nucleus. The threads will represent your chromosomes. The bead will be centromere. You can make the spindle fibre either by thread or a marker.

Activity 2.2: (Pair Activity)

Take cards of process of Mitosis and Meiosis and shuffle them. Distribute one set of cards to each pair of students. Ask students to arrange the cards in sequence. They should arrange Mitosis and Meiosis in a separate sequence. After completion of activity, teacher should facilitate whole class discussion on correct sequence.

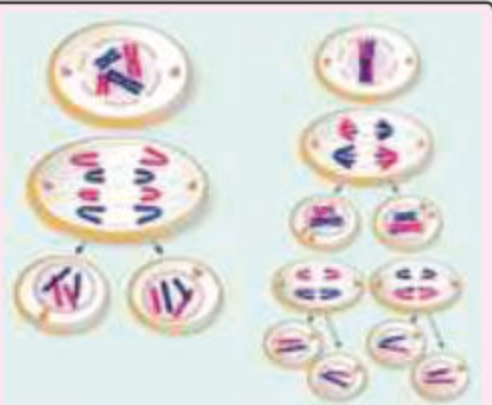


Fig.2.7: Mitosis and Meiosis

BASIS OF HEREDITY

- Identify DNA and chromosomes in the cell diagram.

Do you know what is DNA and where is it located? What does DNA stand for and what is its shape and functions? To understand DNA, let us first explore what is inside the Nuclear material.

Chromosomes

Chromosomes are present inside the nucleus as chromatin network. When cell divides, this chromatin network condenses to

Teacher Note:

Teacher should ask students to draw different stages of Mitosis and Meiosis on separate cards. After completion of the topic they can use these cards for reviewing the topic by Card Sort Activity.

form a typical chromosome consists of two chromatids. The two chromatids are attached at the centre to the same centromere. Chromosomes in pair also called as homologous chromosomes. The number of chromosomes in similar organism is constant.

Genes

The basic physical and functional unit of heredity is called gene. Genes are responsible for the expression of various characteristics. These genes are located on chromosomes. The genes are arranged on chromosomes in a linear order. The number of chromosomes in every organism is fixed.

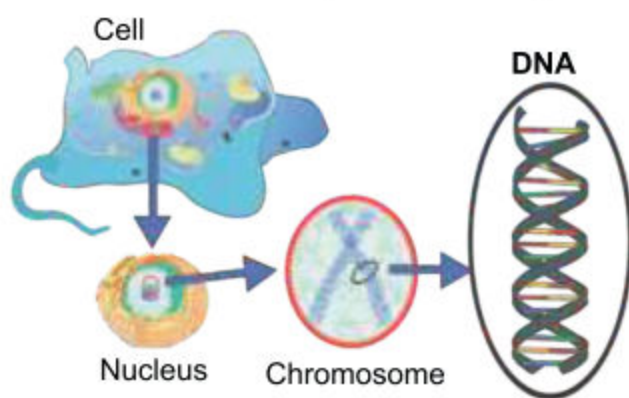


Fig.2.8: Chromosomes

Deoxyribonucleic Acid (DNA)

Chemically a chromosome consists of proteins and nucleic acid. A molecule of DNA consists of two strands linked with each other by braids like a ladder twisted around each other. DNA is a very large molecule and consists of units called nucleotide. A nucleotide consists of phosphate, sugar and base. There are four types of bases, Adenine, Guanine, Thymine and Cytosine. Every living thing has its own distinct DNA. DNA stores genetic information in the sequence of its nucleotide.

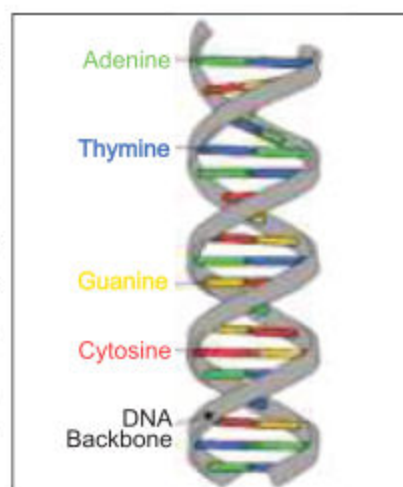


Fig.2.9:DNA

HEREDITY:

- Define heredity and recognize its importance in transferring of characteristics from parents to offsprings.

Have you ever come across a comment that you look exactly like your mother or your hair is curly like your father, how is it possible?

All living things during reproduction pass on their characteristics to their offspring. This is the reason that you may resemble to your

mother or father in some way. Similarly, plants grown from seeds resemble to their parent plants. This transmission of characteristics from parent to offspring is known as inheritance process (Heredity). The characteristics such as eye colour, texture and colour of hair, skin colour, attached or free earlobes are some of the heredity characteristics that pass on from parents to their younger ones.

Activity 2.3:

Exploring Hereditary Characteristics of Family Members

Closely observe some of the physical characteristics mentioned in the table with the members in your family and try to record your observation in the table given below.

Physical traits	You	Mother	Father	Sister	Brother	Paternal Grand parents	Maternal grand parents
Eye colour							
Height							
Texture of hair							
Complexion							

Based on the data in the above table, discuss answers of the following questions with classmates.

- Which physical trait is commonly found in your family?
- Which physical trait was found with your grandparents?
- Did you notice any physical trait which is not found in your brothers or sisters but is found in yourself?

TRANSMISSION OF CHARACTERISTICS:

- Identify the characteristics that can be transferred from parent to offsprings.

As you have already studied about chromosomes and genes, so let us see how different characteristics and traits pass on to offspring by parents. As you know when organisms produce gamete as part of meiosis, the number of chromosome is reduced to half; meaning that the DNA has been reduced to half in the gamete cells. i.e. male cell (n =haploid) and egg cell (n =haploid) will fuse to form a diploid ($2n$) cell called zygote. Thus, zygote is the first cell of an organism from which the new life of an organism begins. In the zygote the complete heredity material is restored, and the new offspring will have some genes from female parent and some genes from the male parent resulting in a offspring with a different genetic make-up.

Inheritable and Non-inheritable Characteristics

Although we are now aware of the phenomenon of genetics, it was a matter of mystery for centuries. How the characteristics are inherited from parents to offsprings? How do we explain the presence or absence of some other characteristics in offsprings? These and similar other questions were commonly asked by the intelligent people of the millennia with no definitive answers.

Gregory Mendel was a pioneer among geneticists who put forward the concept of inheritance of characteristics on traits from parents to offsprings.

EXAMPLES OF INHERITABLE CHARACTERISTICS

- Compare characteristics related to ear and eye colour.



Rolling of tongue



Lobed ear



Lobeless ear

Attached and free earlobe



Colour of eye

Fig.2.10: Inheritable Characteristics

Activity 2.3: Comparing characteristics related to ear and eye colour among your classmates.

Student #	Characteristics	Inherited from			
		<i>Father</i>	<i>Mother</i>	<i>Maternal Grandparents</i>	<i>Paternal grandparents</i>
	Lobed Ear				
	Lobe less Ear				
	Black Eyes				
	Blue Eyes				
	Green Eyes				
	Brown Eyes				

Activity 2.4: Survey your class to observe and find the other types of inherited characteristics among your classmates.

Characteristic	Number of class-fellows
Curly Hair	
Straight Hair	
Ear lobe attached	
Ear lobe free	
Can roll tongue	
Cannot roll tongue	
Fair coloured skin	
Dark coloured skin	
Eye colour	
Hair colour	
Smooth chin	
Cleft chin	

SUMMARY

- DNA, deoxyribonucleic acid resembles a long spiral ladder.
- There are two methods of cell division i.e. Mitosis and Meiosis.
- In Mitosis separation of chromosomes into two identical sets of daughter cells occur.
- Somatic or body cells divide by Mitosis for growth, repairing and development.
- Meiotic cell division is also called as reduction division in which the number of chromosomes are reduced to half during reproduction.
- Sex cells are divided by Meiosis for reproduction.
- The process through which characteristics are transferred from parents to the offspring is called inheritance.
- Children resemble to their parents because they inherit many characteristics from them.

EXERCISE

Q1. Write short answers of the following questions.

- (a) Explain the system in Mitosis cell division.
- (b) What is the purpose of Meiosis?
- (c) What is the purpose of Interphase?
- (d) What does diploid and haploid mean?
- (e) Define following terms:
1. Homologous chromosomes 2. Genes 3. Cytokinesis
- (f) Name two types of cell division and their purpose.
- (g) Differentiate between Mitosis and Meiosis.
- (h) Define heredity and give some examples of hereditary characteristics.
- (i) Define DNA and draw its diagram.

Q2. Fill in the blanks with suitable words.

- (i) Characteristics like attached ear lobes and tongue rolling are _____ from parents.
- (ii) Chromosomes contain _____.
- (iii) The cell divides by two methods: 1. _____ and 2. _____.
- (iv) Prophase means _____.
- (v) Passing on of characteristics or traits from parents to off springs is called _____.

Q.3 Project: Mitosis Flip Book

Instructions: Complete each page to illustrate the changes that take place in a cell during cell division. The first oval (or ovals) in each phase should show the location of the organelles at that stage. Use the extra ovals to show the movement of organelles between stages. Once you have completed all the diagrams, organize this book by stapling different pages in sequence. Flip through your book to view cell division.

Changes during cell division

Mitosis Flip Book
by _____

Interphase



Prophase



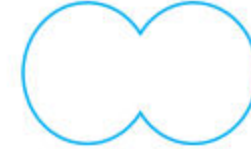
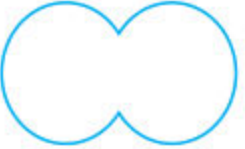
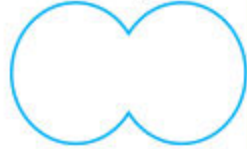
Metaphase



Anaphase



Telophase



Cytokinesis

